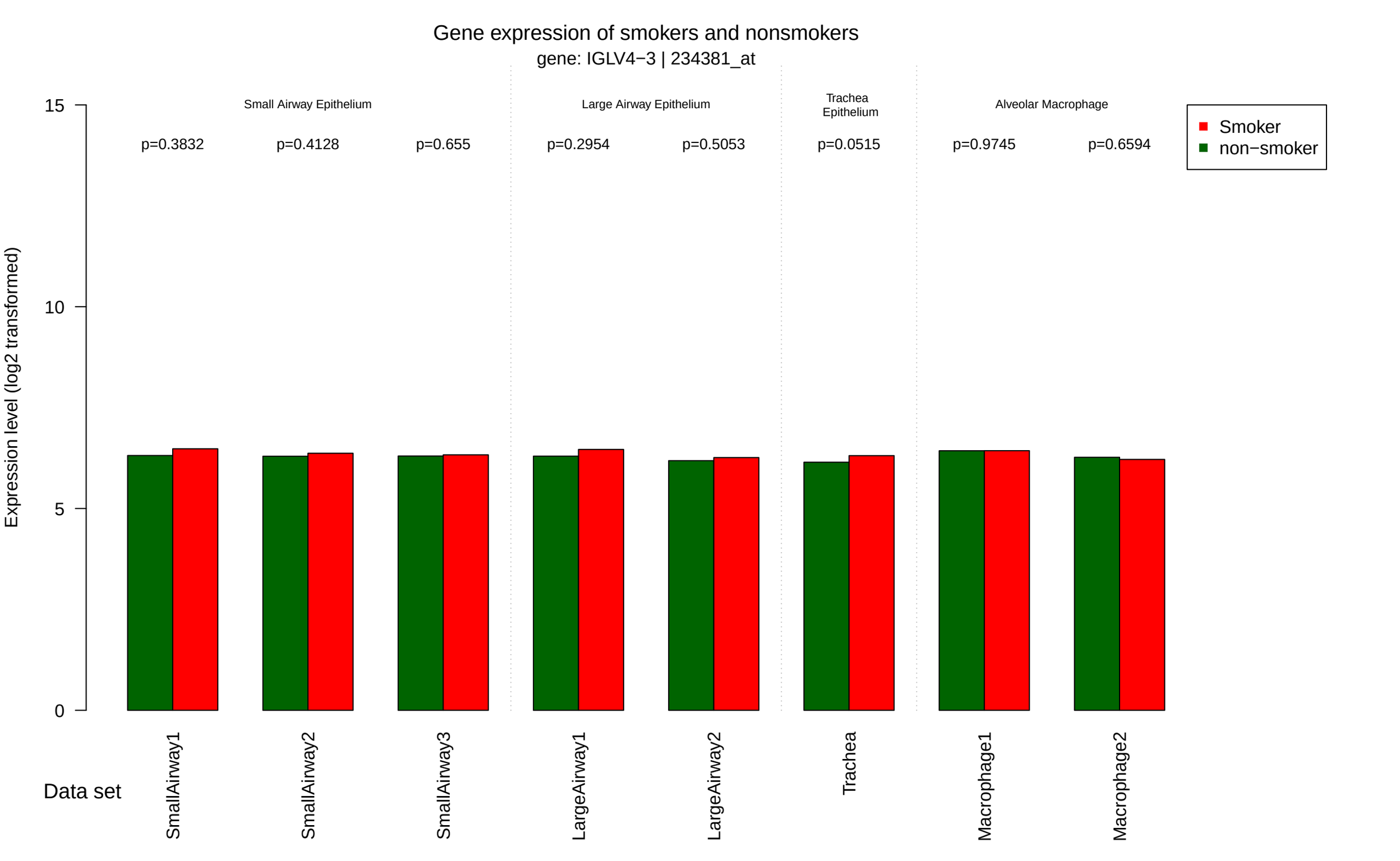
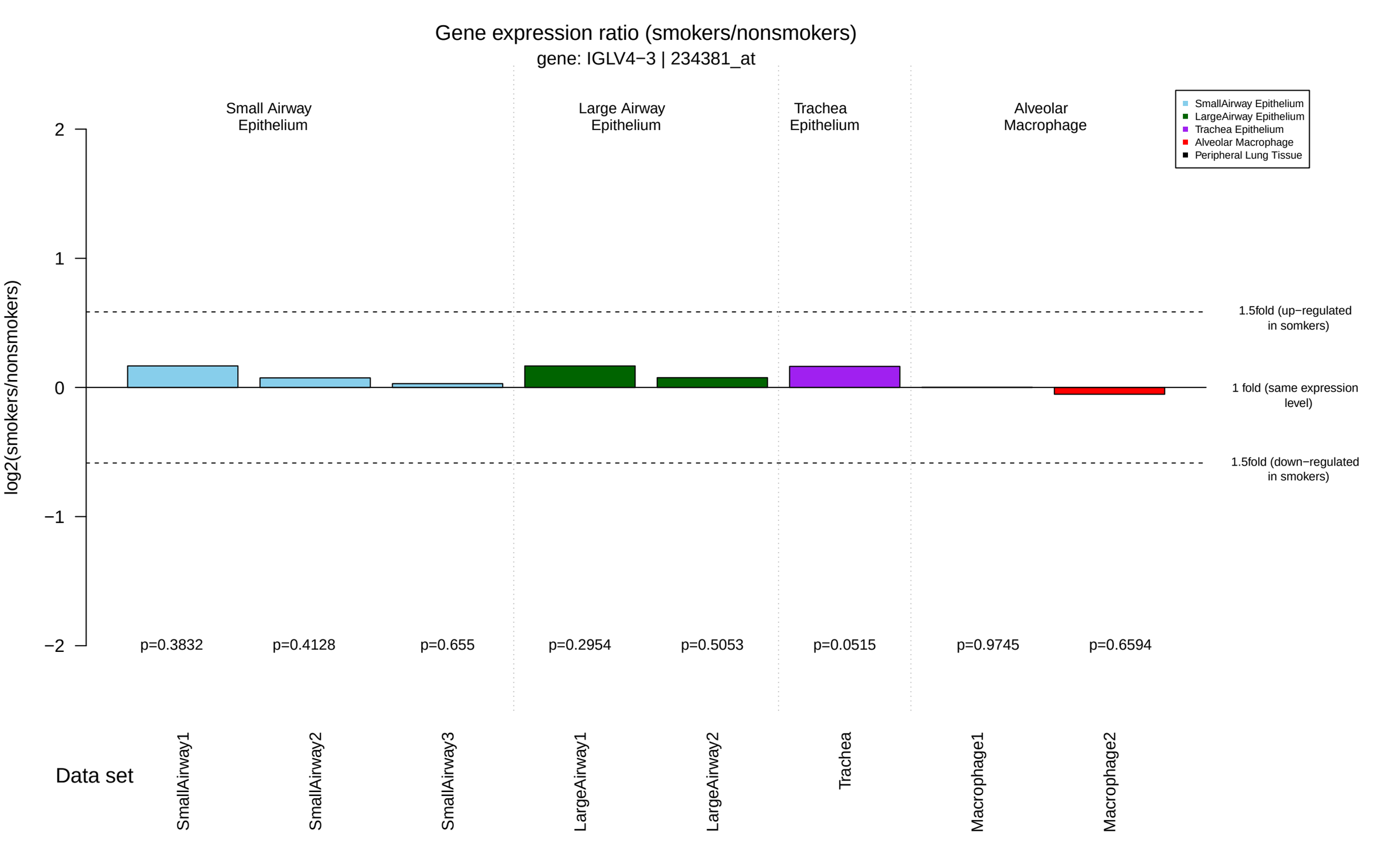
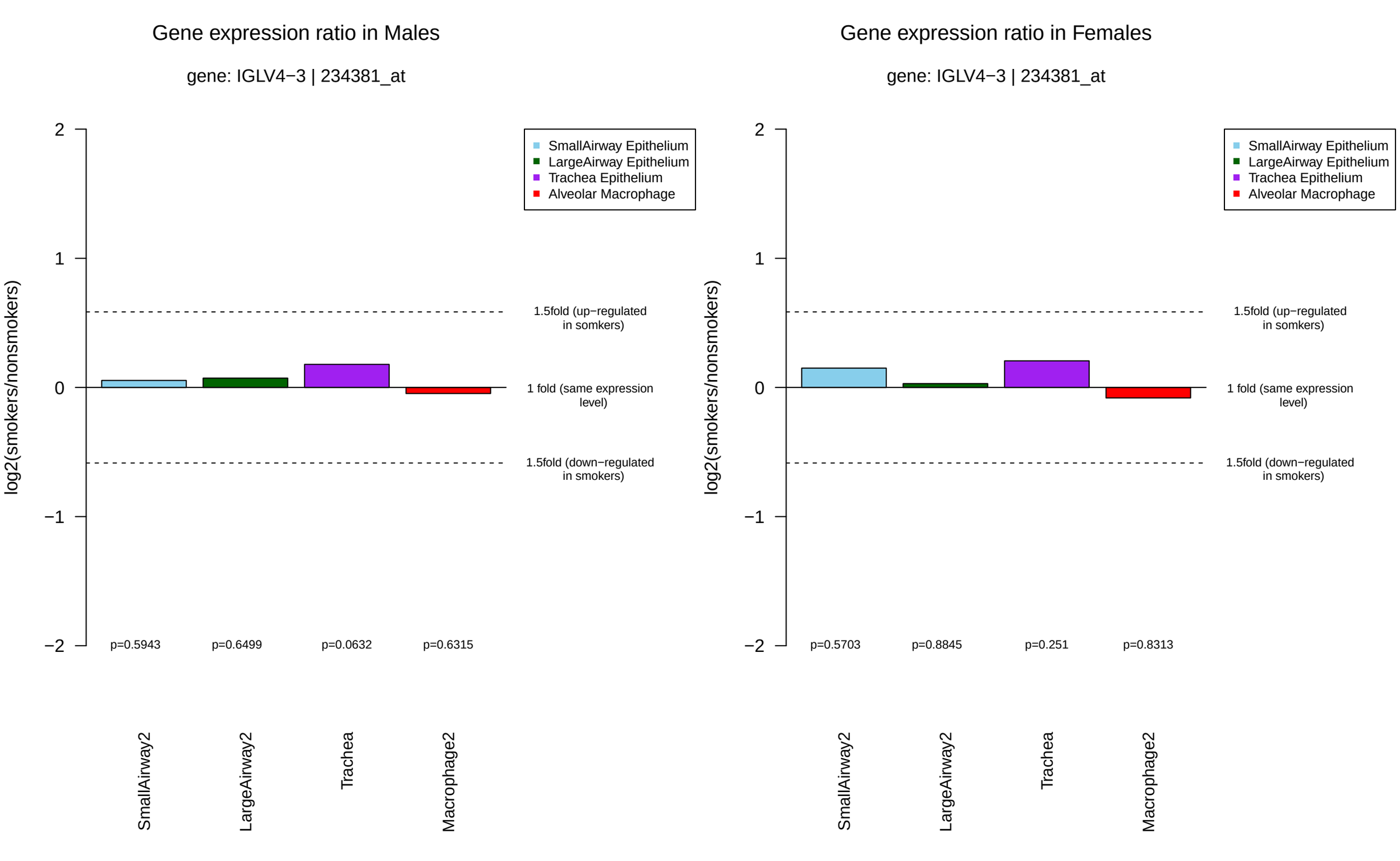
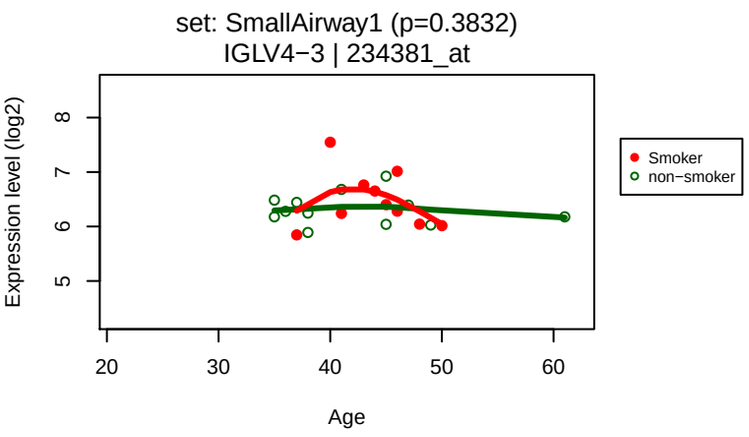
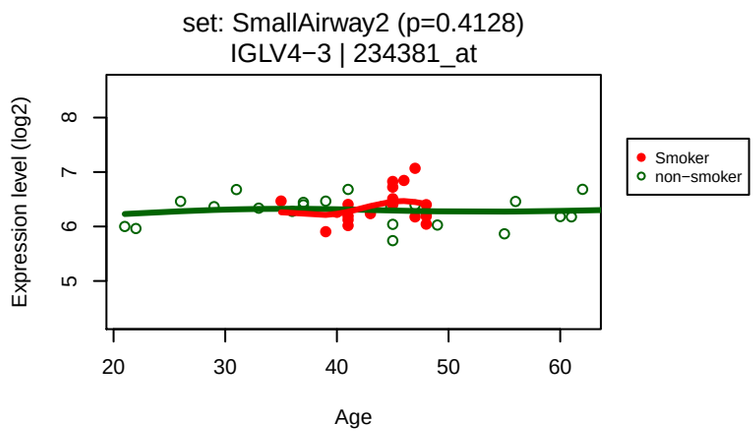
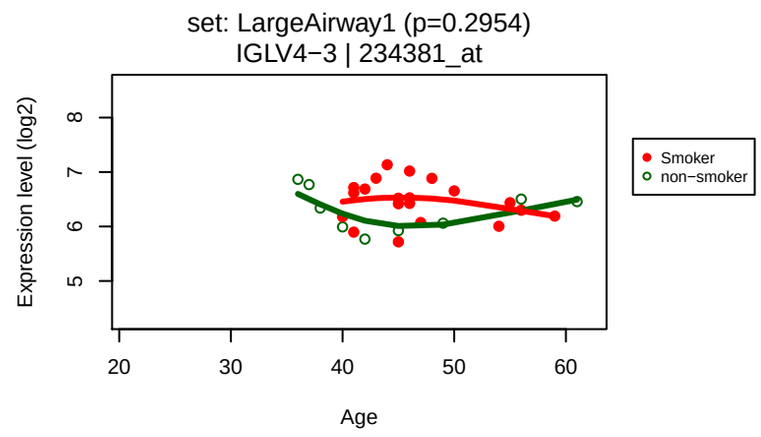

Gene expression of smokers and nonsmokers
gene: IGLV4−3 | 234381_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3832
p=0.4128
p=0.655
p=0.2954
p=0.5053
p=0.0515
p=0.9745
p=0.6594
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

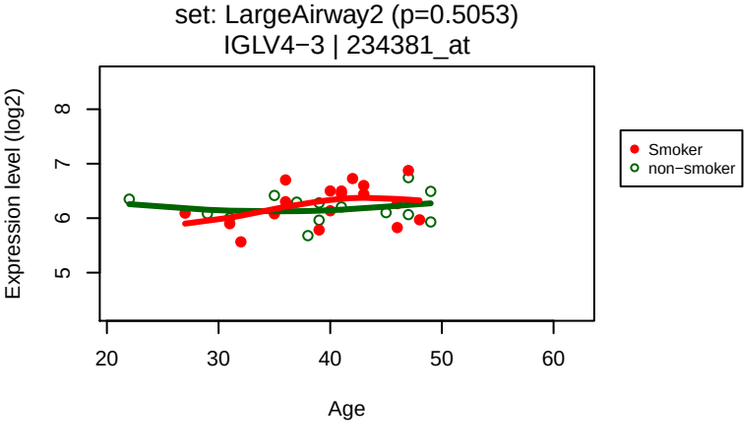

Gene expression ratio (smokers/nonsmokers)
gene: IGLV4−3 | 234381_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3832
p=0.4128
p=0.655
p=0.2954
p=0.5053
p=0.0515
p=0.9745
p=0.6594
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

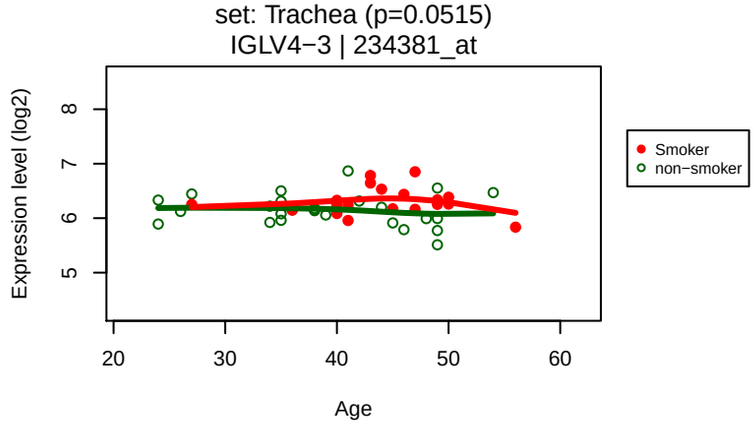

Gene expression ratio in Males
gene: IGLV4−3 | 234381_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5943
p=0.6499
p=0.0632
p=0.6315
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGLV4−3 | 234381_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5703
p=0.8845
p=0.251
p=0.8313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

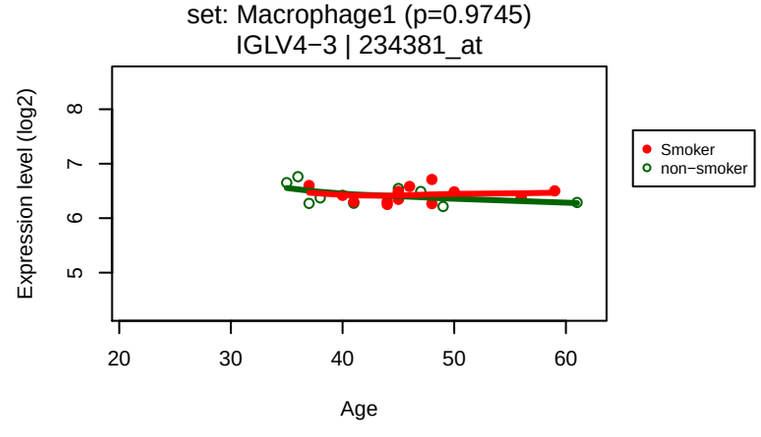

set: SmallAirway1 (p=0.3832)
IGLV4−3 | 234381_at
Expression level (log2)
Age
Smoker
non−smoker

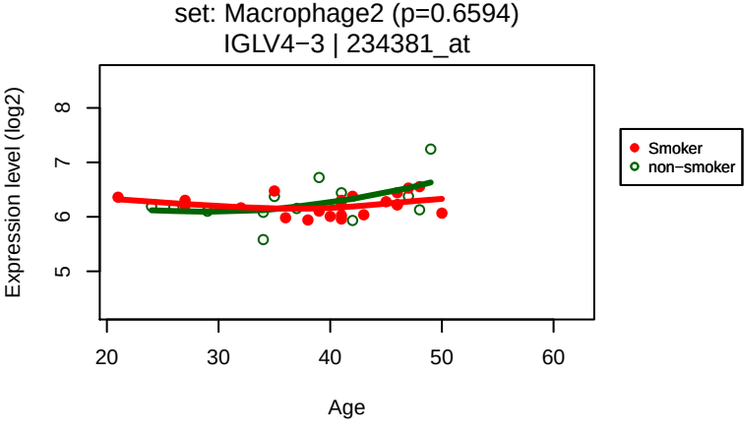

set: SmallAirway2 (p=0.4128)
IGLV4−3 | 234381_at
Expression level (log2)
Age
Smoker
non−smoker

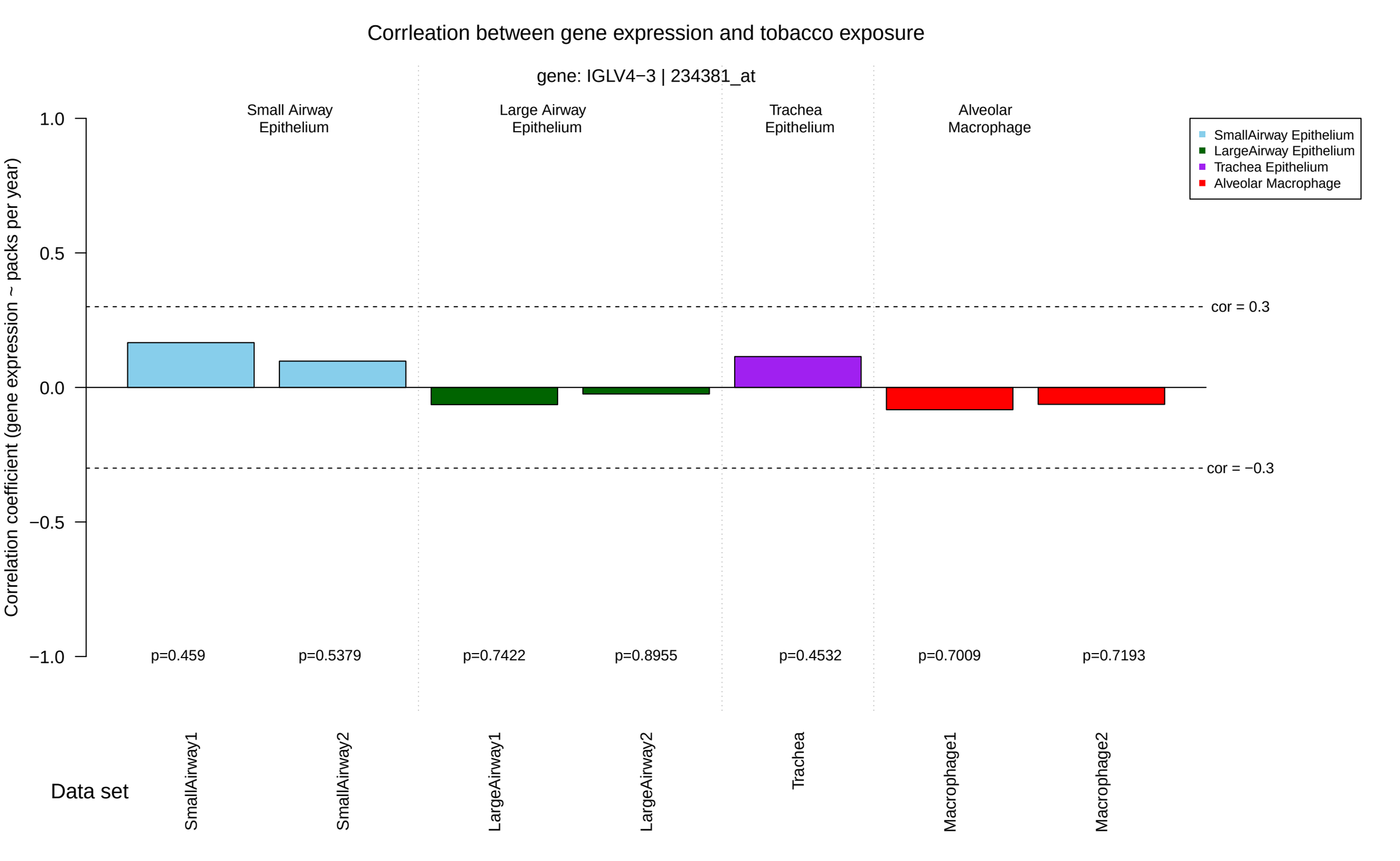

set: LargeAirway1 (p=0.2954)
IGLV4−3 | 234381_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5053)
IGLV4−3 | 234381_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0515)
IGLV4−3 | 234381_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9745)
IGLV4−3 | 234381_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6594)
IGLV4−3 | 234381_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGLV4−3 | 234381_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.459
p=0.5379
p=0.7422
p=0.8955
p=0.4532
p=0.7009
p=0.7193
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2